Kalimantan SAF Optimiser

Technical report, Airbus Challenge 3, Autonomous SG 2026

Wee Zhern Ray, Year 3 Chemical Engineering, National University of Singapore. 31 October 2026.

Live application: weewzr.github.io/airbustrack3ray. Source: github.com/weewzr/airbustrack3ray.

1. Executive summary

A SAF network planner must place about USD 4.7B of plant capital across Kalimantan before engineering starts. This project gives that planner a tool that searches refinery and depot locations and sizes, routes biomass and SAF exactly, and shows the checks behind each number.

On the official inputs of the 31 October update, the best network costs **USD 890.09M per year**, or **USD 742 per tonne SAF**. It uses 6 large refineries, 23 depot modules (21 large, 2 small), 6,000 kt of biomass from 71 sites, and meets all 1,200 kt of demand. On candidate sites the solution is proven within 0.75% of the optimum.

Refinery cost dominates the decision, since transport is only 7.5% of the total. Depots belong at the refinery under the published tariffs, which we prove from the triangle inequality. Moving them to the field costs USD 22.6M more per year and cuts truck-km by 25%, a trade-off we surface for the planner to decide.

Against two rule-of-thumb networks on the same cost basis, the optimised design saves USD 61.3M to USD 76.7M per year (6.4% to 7.9%). These are measured against simple planner rules. They are not a claim about current industry practice.

Reference Frame

Variables and parameters

Symbol	Definition	Units	Value or range
i, j, k, a, s	Biomass site, depot site, refinery site, airport, size (small, large)		80 sites, 4 airports
x_{ij}	Biomass from site i to depot at j	kt/yr	Decision, continuous
u_{jk}	Pre-processed biomass, depot j to refinery k	kt/yr	Decision, continuous
w_{ka}	SAF from refinery k to airport a	kt/yr	Decision, continuous
y_{js}	Depot modules of size s at j	count	Decision, integer
z_{ks}	Refinery of size s open at k	binary	Decision
t_{js}, r_{ks}	Throughput by size at depot and refinery	kt/yr	Decision, continuous
B_i	Biomass available at site i	kt/yr	22.84 to 99.90, sum 6,800
D_a	SAF demand at airport a	kt/yr	300 each
η, ρ	Refinery yield, depot mass retention	t/t	0.20, 1.00
C^D_s, C^R_s	Depot and refinery biomass intake capacity	kt/yr	150, 300; 500, 1,087
F^D_s, F^R_s	Depot and refinery CapEx	USD M	2.25, 4.5; 480, 765
v^D_s, v^R_s	Variable OpEx on biomass in	USD/t	5, 4; 60, 55
τ_1, τ_2, τ_3	Tariff by leg	USD/t/km	0.10, 0.12, 0.08
n, f	CapEx life, fixed OpEx share	yr, 1/yr	20, 0.05
d	Distance between two points	km	Great-circle x 1.3

Equations

The objective is equation (1). Equations (2) to (8) are its constraints. Equation (9) defines distance. Equations (10) to (12) are the solution method. All appear in full in section 6.

Assumptions needed to read them

One-year cost basis with straight-line CapEx and no interest. Biomass mass is taken as stated, with moisture ignored by instruction. All values are deterministic. Masses are annual. Refinery capacity is biomass intake.

2. Airbus problem definition

The brief asks for a SAF supply chain across Kalimantan that minimises OpEx, CapEx and logistics while meeting demand at four airports: Supadio (PNK), Iskandar at Pangkalan Bun (PKN), Sepinggan (BPN) and Juwata (TRK). The chain runs biomass site, pre-processing depot, refinery, airport. The answer must give coordinates, facility sizes and routes. The update of 31 October at 12:45 fixed every open parameter and says it overrides the original where they differ.

3. Kalimantan context and the user

The named user is a SAF network planner at a project developer or fuel supplier. Their job is to recommend sites before front-end engineering. Their usual method is to compare a few hand-drawn layouts in a spreadsheet, which cannot test every site combination and hides why one layout wins. Mistakes here are expensive and persistent, since a misplaced large refinery fixes USD 765M of capital and its haulage for 20 years.

The constraints are physical. Biomass clusters in West, Central and East Kalimantan, the airports sit far apart, and Tarakan is an island reached by barge. The road network is sparse: a sample of 854 site pairs on OpenStreetMap gave a median road distance of 1.9 times great-circle, well above the 1.3 in the brief.

4. Dataset and assumptions

Validation of the official data

Check	Result
Biomass rows and total	80 rows, 6,800.00 kt, matching the brief
Range per site	22.84 to 99.90 kt, inside the stated 20 to 100 kt
Null cells, duplicate ids	0, 0
kt and tonnes columns agree	Yes, on every row
Airports	4 rows, 300 kt each, 1,200 kt total
Refinery cost scaling	765 / 480 equals $(1{,}087 / 500)^{0.60}$, the six-tenths rule
Depot unit CapEx	USD 15k per kt for both sizes, so large wins only on variable cost
Coastline check	13 of 80 points, 1,153 kt, fall off the Natural Earth 10 m coast

We treat the 13 off-coast points as genuine biomass sources, since the coordinates are official, but they cannot host a refinery because the brief requires land. The first version of the solver missed this. The independent audit caught a plant placed on one of those points, and the candidate set was corrected.

Assumption register

Input	Value	Status	Source
Capacity basis	Biomass intake	Official	Update, 12:45
Yield	0.20 t SAF per t biomass	Official	Update, 12:45
Depot mass loss	0%	Official	Update, 12:45
Demand	300 kt per airport	Official	challenge3_airports_4.csv
Costs	CapEx / 20, fixed OpEx 5%, variable per tonne	Official	challenge3_parameters.xlsx
Distance	Great-circle x 1.3	Official	Update, 12:45
Tarakan leg	Barge at 0.08 USD/t/km	Official	Update, 12:45
Refinery candidates	On-land sites plus PNK, PKN, BPN	Ours	Tarakan excluded, as biomass cannot reach the island
One refinery per site	At most one	Ours	Depot modules may share a site
Field depot radius	60 km	Ours	Used only in the field design
Real uplift	About 100 kt/yr	Public	Reality check only, see section 14

5. Literature review

Six papers shaped the formulation. Ng and Maravelias [3] build a multi-period MILP with variable depot and biorefinery locations, and report that sites near a biorefinery ship directly while distant sites justify a depot. That pattern depends on densification lowering later transport cost, which the brief's tariffs do not offer. Their example took 3,405 s to reach a 0.43% gap with 431 binaries, a warning about scale. Ng et al. [2] stress accurate distances and show that pruning reduces 557 candidate depots and over three million arcs to 378 depots and 80,000 arcs, which supports candidate reduction.

Huang et al. [5] compare ATJ, FT and HTL on corn stover. They report that large centralised plants win through economies of scale, which matches our result. Li et al. [4] show that assuming fixed yields overstates maximum SAF supply by 14.15% and net revenue by 18.27% when ash varies, so yield deserves scrutiny. Hernandez-Araujo et al. [6] couple supply chain and process models for bagasse-based ATJ, and Castillo-Villar et al. [7] use machine learning to pick facility configurations for a county, depot, biorefinery, airport network, which is the same four-tier shape as this problem.

We take the four-tier mass balance from [7], the strong linking constraints from the facility location literature, and the candidate-reduction idea from [2]. We did not combine their cost data, because the update fixes all costs. Course material from CN4252 [8, 9] informed the annualisation logic and the assumption register.

6. Mathematical formulation

Minimise annual cost:

$$\min\ (\tfrac{1}{n}+f)\Big[\sum F_s^D y_{js}+\sum F_s^R z_{ks}\Big]+\sum v_s^D t_{js}+\sum v_s^R r_{ks}+\sum \tau_1 d_{ij}x_{ij}+\sum \tau_2 d_{jk}u_{jk}+\sum \tau_3 d_{ka}w_{ka} \quad (1)$$

Subject to biomass availability, depot balance and capacity, depot mass retention, refinery balance and capacity, conversion, demand, and domains:

$$\sum_j x_{ij} \le B_i \quad \forall i \quad (2)$$

$$\sum_i x_{ij} = \sum_s t_{js}, \qquad t_{js} \le C_s^D\, y_{js} \quad \forall j,s \quad (3)$$

$$\sum_k u_{jk} = \rho \sum_i x_{ij} \quad \forall j \quad (4)$$

$$\sum_j u_{jk} = \sum_s r_{ks}, \qquad r_{ks} \le C_s^R\, z_{ks} \quad \forall k,s \quad (5)$$

$$\sum_a w_{ka} = \eta \sum_s r_{ks} \quad \forall k \quad (6)$$

$$\sum_k w_{ka} \geq D_a \quad \forall a \tag{7}$$

$$\sum_s z_{ks} \leq 1, \quad z_{ks} \in \{0,1\}, \quad y_{js} \in \mathbb{Z}_{\geq 0}, \quad x, u, w, t, r \geq 0 \tag{8}$$

$$d_{pq} = 1.3 \times 2R \arcsin \sqrt{\sin^2 \tfrac{\Delta\phi}{2} + \cos\phi_p \cos\phi_q \sin^2 \tfrac{\Delta\lambda}{2}} \tag{9}$$

Truck size

The tariff is per tonne-km and the same for both truck sizes, so truck size cannot enter the objective. We report trip counts only, and we do not model trucks as a decision.

Why depots sit at refineries

Take biomass moving from site i through a depot at j to a refinery at k. Its transport cost per tonne is $\tau_1 d_{ij} + \rho\tau_2 d_{jk}$. If $\rho\tau_2$ is at least τ_1, this is at least $\tau_1(d_{ij} + d_{jk})$, which by the triangle inequality is at least $\tau_1 d_{ik}$. Moving the depot to k therefore never raises transport cost, and it keeps the same modules at the same cost. With $\rho = 1$ and $\tau_2 = 0.12$ against $\tau_1 = 0.10$, co-location is optimal. The result reverses below a retention of 0.833, which is why the brief's instruction to ignore moisture loss matters. A real depot that dries wet biomass would earn its place in the field.

7. Algorithm and solver selection

Method	Assessment	Decision
General MILP solver (HiGHS)	Reliable with a proven bound, but C++ and heavy to run live in a browser	Kept as the independent check
Purpose-built Lagrangian branch-and-bound in Rust	Exploits problem structure, compiles to WebAssembly, gives a true bound	Chosen engine
MINLP with free coordinates	Non-convex, no practical global proof at this size	Replaced by a two-step method
Clustering plus MILP	Fast but depends on arbitrary cluster counts	Used only as a starting design
Pure metaheuristic	No bound, so no honest statement of quality	Rejected

Measurements drove this choice. In a timed run of a general MILP on an earlier version of this problem, 99.1% of runtime sat inside the solver, so a language change alone could not help. The gain came from using the structure. Relaxing supply and demand with multipliers u and λ splits the model into one subproblem per site:

$$L(u,\lambda) = \sum_a \lambda_a E_a - \sum_i u_i B_i + \min_z \sum_{k,s} z_{ks}\,(G_s + V_{ks}(u,\lambda)), \qquad \sum_{k,s} C_s^R z_{ks} \geq \sum_a E_a \tag{10}$$

Here E_a is the biomass equivalent of demand, G_s the annual fixed cost of a plant and V_{ks} the best value of site k, found by pairing the cheapest inputs with the dearest outputs. Any multipliers give a valid lower bound, and subgradient steps tighten it. Branching on open and closed sites closes the gap, and each design found is priced exactly by min-cost flow. Sites may be anywhere on land, so a final step alternates a location move and a re-allocation [10], where each plant moves by Weiszfeld iteration [11]:

$$\mathbf{p}^{(t+1)} = \left(\sum_m \frac{\omega_m \mathbf{p}_m}{\|\mathbf{p}^{(t)} - \mathbf{p}_m\|}\right)\left(\sum_m \frac{\omega_m}{\|\mathbf{p}^{(t)} - \mathbf{p}_m\|}\right)^{-1}, \qquad \omega_m = \tau \times \mathrm{flow}_m \tag{11}$$

A land mask rejects offshore moves. This step converges to a local optimum and carries no global proof.

8. System architecture

The engine is a Rust library covering data, instance building, min-cost flow, search, refinement, audit, baselines, diagnostics and decision support, with 25 automated tests. A command-line tool runs batches and exports the model. A WebAssembly build of the same library runs inside the browser in a Web Worker, so the deployed site on GitHub Pages needs no server, account or API key. The page uses Leaflet for the map and

Chart.js for charts. Python appears only in supporting scripts for data fetching, the HiGHS check and these documents.

9. Optimisation results

Measure	Value
Total annual cost	USD 890.09M
Cost per tonne SAF	USD 742
Candidate-site cost	USD 890.32M
Proven lower bound, candidate sites	USD 883.60M
Proven gap	0.75%
Search effort	6,367 nodes, 478,813 bound iterations, 60 s
Biomass used	6,000 kt from 71 sites
SAF delivered	1,200 kt, 100% of demand
HiGHS cost, same model	USD 890.32M (Time limit reached)
HiGHS lower bound	USD 866.74M
HiGHS gap and time limit	2.65%, 480 s

10. Facility location results

Refinery	Size	Latitude	Longitude	Biomass kt	SAF kt	Used	Nearest site
R1	large	-2.0113	116.0324	807	161	74.2 %	24 km from KAL_SUP_24 (Paser, East Kalimantan)
R2	large	-1.7167	110.0022	1,087	217	100.0 %	1 km from KAL_SUP_41 (Ketapang, West Kalimantan)
R3	large	-0.3788	116.9294	986	197	90.7 %	4 km from KAL_SUP_44 (Kutai Kartanegara, East Kalimantan)
R4	large	0.2508	110.6944	1,087	217	100.0 %	3 km from KAL_SUP_46 (Sanggau, West Kalimantan)
R5	large	-2.5351	111.6833	1,087	217	100.0 %	2 km from KAL_SUP_65 (Kotawaringin Barat, Central Kalimantan)
R6	large	-2.4215	112.9479	947	189	87.1 %	1 km from KAL_SUP_71 (Kotawaringin Timur, Central Kalimantan)

Depot modules at each refinery: R1: 3 large; R2: 4 large; R3: 1 small, 3 large; R4: 4 large; R5: 4 large; R6: 1 small, 3 large.

Continuous refinement saved only USD 0.23M per year. Biomass sites sit in tight clusters, so the best candidate site is already close to the geometric median. Restricting plants to candidate points therefore costs almost nothing here.

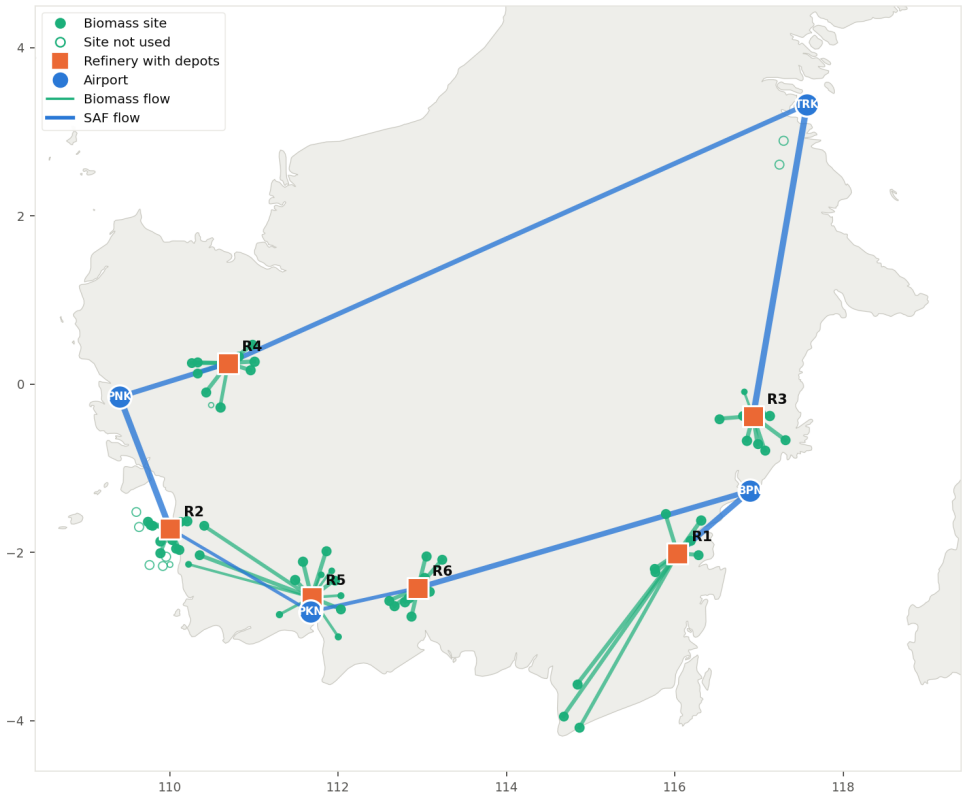


Figure 1. Optimised network. Refineries sit inside the biomass clusters, with depots at each refinery. Lines are schematic, and the last leg to Tarakan is by barge.

11. Transport and material flows

From	To	SAF kt	Distance km	USD M/yr	Mode
R1	BPN	161.3	164	2.12	truck
R2	PNK	185.5	242	3.60	truck
R2	PKN	31.9	280	0.72	truck
R3	TRK	197.1	544	8.57	truck then barge
R4	PNK	114.5	195	1.79	truck
R4	TRK	102.9	1,088	8.95	truck then barge
R5	PKN	217.4	25	0.43	truck
R6	PKN	50.7	189	0.76	truck
R6	BPN	138.7	594	6.59	truck

Sites unused: KAL_SUP_03, KAL_SUP_18, KAL_SUP_22, KAL_SUP_31, KAL_SUP_40, KAL_SUP_50, KAL_SUP_69, KAL_SUP_70, KAL_SUP_80. These account for the 800 kt not needed. Mean biomass haul is 56 km and mean SAF haul 349 km. Biomass is five times heavier per tonne of SAF, so plants stay close to it and SAF travels further. Delivery to Tarakan is the longest and the sole barge leg.

Leg	Tonnes	Small truck	Trips	Large truck	Trips
Farm to depot	6,000,000	3 t	2,000,001	9 t	666,667
Depot to refinery	6,000,000	18 t	333,334	30 t	200,000
Refinery to airport	1,200,000	16 t	75,000	32 t	37,500

12. Cost analysis

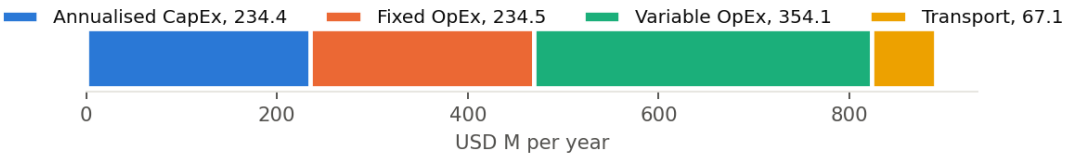


Figure 2. Annual cost by component, official base case.

Component	USD M/yr	Share
Annualised CapEx	234.4	26.3%
Fixed OpEx	234.5	26.3%
Variable OpEx, refineries	330.0	37.1%
Variable OpEx, depots	24.1	2.7%
Transport, site to depot	33.5	3.8%
Transport, refinery to airport	33.5	3.8%
Total	890.1	100%

Total CapEx is USD 4,689M, of which refineries are 97.9%. The objective excludes biomass purchase, as the update defines it, so USD 742/t is a delivered cost before feedstock. Readers should not compare it directly with a market SAF price.

13. Baseline comparison

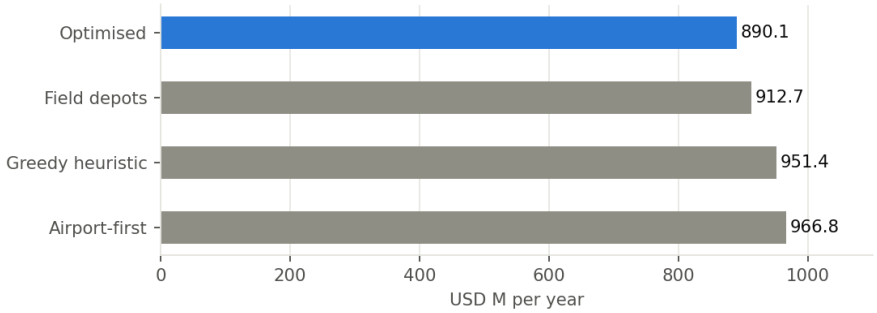


Figure 3. Annual cost on identical demand, distances and cost basis.

Design	USD M/yr	Extra cost vs optimised	Refineries	Depots	Truck-km, M/yr
Optimised	890.1	n/a	6	23	100.7
Field depots, 60 km radius	912.7	22.6 (2.5%)	6	25	75.3
Greedy: richest sites, farthest-first plants	951.4	61.3 (6.4%)	6	20	116.7
Airport-first: plants at PNK, PKN, BPN	966.8	76.7 (7.9%)	6	20	153.2

All baselines passed the same audit. The field design also keeps the optimiser's refinery sites, so its depots are the only source of its extra cost. Truck-km assume large trucks and an empty return. On that measure the field design is the better one, which a tonne-km tariff cannot see.

14. Scenario and sensitivity analysis

Scenario	Status	USD M/yr	USD/t	Ref	Dep	Gap
Official base	feasible_gap	890.1	742	6	23	0.75%
High demand, +10%	feasible_gap	986.0	747	7	25	0.85%
High transport, +50%	feasible_gap	923.6	770	6	23	0.84%
All 6,800 kt, 340 kt per airport	feasible_gap	1015.1	746	7	25	2.66%
Field depots	heuristic	912.7	761	6	25	0.75%
OSRM road distances	feasible_gap	917.4	765	6	23	0.60%
Real uplift at 5% blend	optimal	50.0	10,031	1	1	0.43%

High demand forces a small plant into the fleet because six large plants take only 6,522 kt. Raising tariffs by 50% adds USD 33.5M and leaves the fleet at six plants. Figure 4 shows why: refinery terms dominate, and yield matters most, since it sets how much biomass and capacity are needed at all.

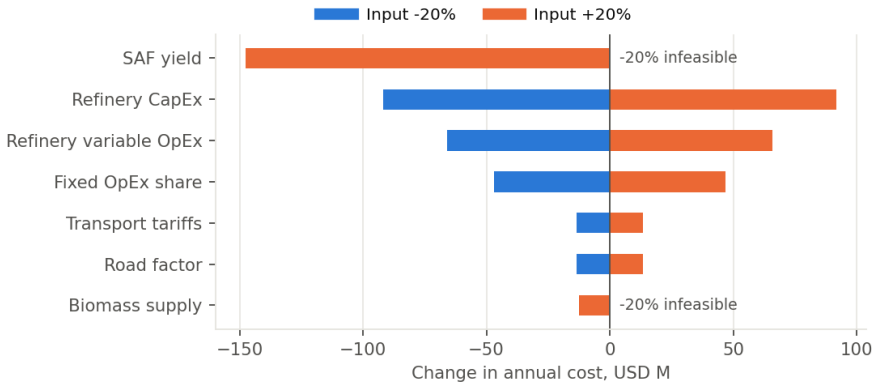


Figure 4. One-at-a-time changes of 20% around the base case. Yield and biomass at -20% are infeasible.

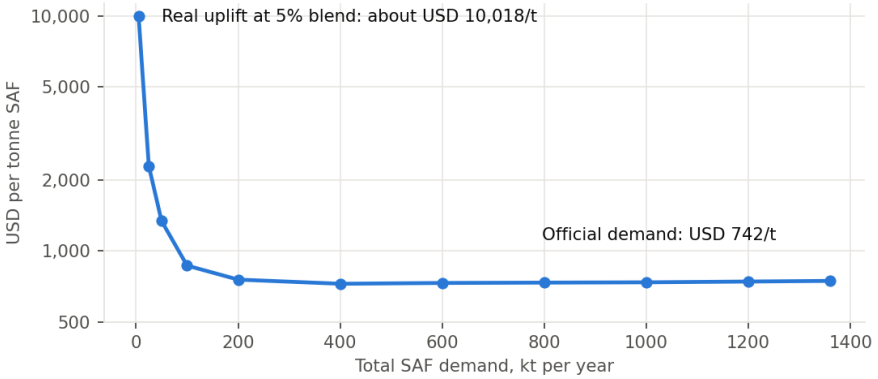


Figure 5. Cost per tonne by demand level. Each point is a full solve.

Reality check on demand

The official demand of 1,200 kt is far above what these airports use today. We estimate real jet uplift at about 100 kt per year, from Pertamina volumes at BPN and PNK [13, 14] and DGCA and BPS passenger counts [15], so official demand is about 12 times total current jet use. Indonesia's 1% SAF mandate from 2027 covers international departures from Jakarta and Bali only [16], so we applied Singapore's 3 to 5% target for 2030 [17] as a stand-in. A 5% blend gives about 5 kt SAF, which the model serves with one small refinery at 5% utilisation at USD 10,031/t. The official figure should therefore be read as a future or regional hub volume. Cost per tonne only settles once plants run near capacity, as Figure 5 shows.

Reality check on distance

With OSRM road distances in place of great-circle x 1.3, annual cost rises by USD 27.1M to USD 917.4M, a 3.0% change, and the fleet size holds. Transport's low share limits the effect.

Where the model fails

Case	Result	What the planner sees
Demand +20%	Infeasible	Demand of 1440 kt SAF needs 7200 kt biomass at yield 0.20, but only 6800 kt exists. This feedstock can make at most 1360 kt SAF. Cut demand by 5.6%, or raise yield to at least 0.212.
Yield 0.15	Infeasible	Demand of 1200 kt SAF needs 8000 kt biomass at yield 0.15, but only 6800 kt exists. This feedstock can make at most 1020 kt SAF. Cut demand by 15.0%, or raise yield to at least 0.176.
At most 5 refineries	Infeasible	5 large refineries take 5435 kt biomass, but this scenario needs 6000 kt. Allow at least 6 refineries.

Demand can rise by at most 13.3% before feedstock runs out. Yield below 0.176 makes 1,200 kt unreachable. The update allows yields of 15 to 25%, so the bottom of that range cannot meet official demand.

15. Decision support method

The rubric asks for language models to handle explanation and for solvers to handle optimisation. We went further and left language models out, so no model can invent a figure and no API key is needed. A question routes by rule to an intent. Each intent reads named fields from a solver result and lists them under its answer. A what-if request becomes a parameter change and a real solver run. A refinery explanation includes a counterfactual solve with nearby sites closed. Questions about emissions, price or law are declined with a reason, because the model holds cost, mass and capacity only. The trade-off is that the system understands a narrower range of phrasing than a language model would.

16. Validation and limitations

A separate audit module recomputes every balance and the objective from flow records, using its own distance calculation. All 11 checks pass on the base case. Automated tests cover input data, capacities, mass balance, yield, demand, objective arithmetic, recalculation, infeasible cases and tampering, and all 25 pass. Tampering tests confirm the audit fails when a flow or objective is altered.

As a second opinion, the Rust tool exports the candidate-site model as a plain LP file and HiGHS 1.15.1 solves it separately. HiGHS reaches USD 890.3215M with a lower bound of USD 866.74M (2.65% gap at its 480 s limit). The Rust cost on the same candidate sites is USD 890.3215M with a bound of USD 883.60M (0.75% gap in 60 s). Two separate codes reached the same cost, which is strong evidence that the formulation and the search agree.

A tighter bound is only useful if it is valid, so we tested it on 5 smaller demand levels that HiGHS can close. In each case the Rust bound must not exceed the HiGHS optimum.

Total demand, kt	Rust cost	Rust bound	HiGHS cost	HiGHS bound	Bound valid
20	55.699	55.524	55.699	55.699	Yes
50	67.126	67.051	67.126	67.126	Yes
100	86.729	86.579	86.729	86.729	Yes
200	151.178	151.003	151.178	151.178	Yes
300	231.788	231.325	231.788	231.788	Yes

- The gap is proven for the candidate-site problem. Refined coordinates are a local optimum.
- The model is deterministic and annual, with no seasonality, storage or supply risk.
- Biomass price is outside the objective by instruction. Including it would favour the same sites only if prices are uniform.
- Co-located depots follow from zero mass loss and the published tariffs. Real wet feedstock would change this, as section 6 shows.

- The field design is a heuristic with no optimality proof, so its USD 22.6M premium is an upper estimate.
- Candidate sites are the biomass points and three airports. Road access, land tenure and water supply are not checked.
- Real uplift for PKN and TRK is scaled from passengers, and the PNK volume is reported for West Kalimantan as a whole.

17. Conclusions

A six-plant network at USD 890M per year meets official demand, with the result checked by an independent audit and a second solver. The strongest levers are refinery cost and yield, and transport matters much less than a logistics framing suggests. The tool's value lies in letting a planner test these questions in seconds and see where an idea fails. The next step for a real project would be to replace the tariff assumption with site-specific moisture data, because that single change decides whether depots belong in the field.

18. References

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